Jingxuan (Jasper) Wu

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Research Interest

My research aims to build controllable generative systems through the integration of diffusion/flow-based models and multi-agent systems: (1) **Generative Models**—enabling diverse, fine-grained controllable generation under limited supervision through both training-free inference-time control and RL-based training; (2) **Agentic Systems**—designing multi-agent systems that interact, coordinate, and adapt through feedback.

Education

The University of North Carolina at Chapel Hill — Aug. 2025 – Dec. 2026 (Expected)
M.S. in Statistics — *Chapel Hill, NC, USA*

The Chinese University of Hong Kong, Shenzhen — Sep. 2021 – May 2025
B.S. (Hons) in Data Science and Big Data Technology — *Shenzhen, China*

Selected Publications

(* indicates equal contribution)

Conference & Workshop Papers

[C1] **Jingxuan Wu**, Zhenglin Wan, Xingrui Yu, Yuzhe Yang, Bo An, and Ivor Tsang. **"OSCAR: Orthogonal Stochastic Control for Alignment-Respecting Diversity in Flow Matching"**. 2025.
ICML 2026.

[C2] Zhenglin Wan, **Jingxuan Wu**, Xingrui Yu, Chubin Zhang, Mingcong Lei, Bo An, and Ivor Tsang. **"FM-IRL: Flow-Matching for Reward Modeling and Policy Regularization in Reinforcement Learning"**. 2025.
ICML 2026.

Journal Papers

[J1] Chi Li, Jianfeng Mao, Lingyi Li, **Wu, Jingxuan**, Lianmin Zhang, Jianyu Zhu, and Zibin Pan. **"Flight delay propagation modeling: Data, Methods, and Future opportunities"**. In: *Transportation Research Part E: Logistics and Transportation Review* 185 (2024), p. 103525.

Preprints & Technical Reports

[P1] **Jingxuan Wu**, Zhenglin Wan, Xingrui Yu, Yuzhe Yang, Yiqiao Huang, Ivor Tsang, and Yang You. **"Time-Annealed Perturbation Sampling: Diverse Generation for Diffusion Language Models"**. 2026.

[P2] Wang, Hanzhao*, **Wu, Jingxuan***, Pan, Yu, Li, Yumeng, Wang, Yansong, Liu, Helang, Wang, Fuqiang, and Chen, Guanting. **"LLM-Powered Predictive Decision-Making for Sustainable Data Center Operations"**. OpenReview. 2024.
In Preparation for *Manufacturing & Service Operations Management*.

Research Experience

Statistics & Operations Research, University of North Carolina, Chapel Hill — Jan. 2024 – Present
Advisors: Prof. Guanting Chen, Prof. Xiaocheng Li, Prof. Wenjia Ba — *Chapel Hill, NC, USA*

• **Adaptive Preference Learning for Generative Design via Hierarchical Proxy-Human Agents**: Built an end-to-end experimental pipeline for Diffusion Language Models, integrating backbones, inference-time interventions, decoding baselines, and automatic evaluation of semantic diversity and generation quality.
• **Resource Allocation Optimization for GPUs using LLMs**: Developed an LLM-driven GPU scheduling pipeline that predicts runtime/energy from code embeddings, models arrivals with a non-homogeneous Poisson process, and evaluates an online allocator on datacenter traces to improve performance.

A*STAR Centre for Frontier AI Research May. 2025 – Present
Advisors: Prof. Xingrui Yu, Prof. Ivor Tsang, Prof. Bo An *Singapore*

• **Time-Annealed Perturbation Sampling for Diffusion Language Models (TAPS)**: Proposed a training-free, scalable inference-time perturbation scheme with time-dependent context annealing and quality-preserving rescaling to boost generation diversity in Diffusion Language Models.
• **Guidance Mechanisms to Enhance Diversity in Flow Matching**: Developed a training-free inference-time control framework (OSCAR) that improves set-level diversity while preserving quality/alignment via feature-space volume maximization and orthogonal stochastic control injected into the base flow.
• **Flow-Matching Rewards & Policy Regularization**: Introduced a teacher–student FM-IRL framework where a flow-matching teacher provides reward shaping and policy regularization, enabling stable online updates and improved efficiency/robustness under limited or suboptimal demonstrations.

School of Data Science, CUHK-Shenzhen Dec. 2023 – Nov. 2024
Advisors: Prof. Jianfeng Mao *Shenzhen, China*

• **Research on Network-wide Delay Propagation Dynamics Prediction**: Modeled network-wide flight delay propagation with a heterogeneous time-varying SIS process and an adaptive graph learning framework (GAT-based) to learn infection/recovery dynamics, achieving improved multi-horizon delay-status forecasting.
• **Study on Flight Delay Propagation Modeling**: Conducted a structured review of 40+ studies on airport delay propagation across statistical/econometric/queueing paradigms, synthesizing assumptions, strengths/limitations, and data-scale considerations into a comprehensive survey manuscript.

Projects

CaveAgent | Python, LLM, AST, LiteLLM, Asyncio, Security Validation Oct. 2025 – Jan. 2026
A tool-augmented agent framework enabling stateful runtime management and code-based function calling.
• Contributed to the development of the persistent Python runtime modules, supporting seamless object injection and retrieval across multi-turn LLM interactions.
• Implemented specific AST-based security rules to validate and sanitize LLM-generated code before execution.
• Developed discovery and registry mechanisms for the Agent Skills open standard, allowing modular injection of functions and types into the runtime.
• Participated in benchmarking and performance evaluation across multiple LLM backends (OpenAI, Claude, Gemini) to validate framework robustness.

References available upon request.